

TOPIC TWO
SUB TOPIC 5: MINERALS MICROMOLECULE

Lecture 4

MINERALS

MICROMOLECULE

Learning Objectives

- Define Mineral in chemical terms
- Classify Minerals into two major groups with examples of each group
- List the **Major minerals** Minerals of biological importance and learn their properties
- List the **Trace elements** Minerals of biological importance and learn their properties
- Explain biological importance/functions of each Minerals
- Describe the diseases/D isorder associated with Minerals

Introduction

- **Required for** : growth, repair, and regulation of vital body functions. Two major groups:
- **Major minerals** : calcium, phosphorus, sodium, potassium, magnesium;
- **Trace elements** : iron, iodine, flourine, zinc, copper, cobalt, Chromium, Manganese etc. (requirement-- less than few mg/day.)

IRON

Causes of iron deficiency:

- • Storage depletion
- • Reduced dietary intake.
- • Malnutrition
- • Hemolysis
- • Children who are on milk diet only are prone to iron deficiency.
- • Chronic bleeding, irregular menstrual cycles
- • Peptic ulcer, piles
- • Hook worm infection
- • Repeated malarial infections.

CALCIUM:

- **Absorption:** It is influenced by
 - Acidic pH solubilizes Calcium salts, promote absorption.
 - High protein diet favors absorption
 - Certain plant products, high fiber diet, oxalates interfere with absorption.
 - Vitamin D promotes absorption.
 - PTH, Calcitonin favors absorption while Glucocorticoids decrease intestinal transport.
 - Normal blood concentration is critically maintained at 9-11 mg %.

Functions:

- • Calcification of bones and teeth. Bone formation requires Calcium continuously.
- • It is important for blood coagulation
- • Neuromuscular transmission.
- • Muscle contraction
- • Acts as secondary messenger in hormone action.

Clinical conditions:

- Hyper- calcemia; may be due to hyper parathyroidism, endocrine causes, renal failure and
- malignancies. Hypo- calcemia (below 8.5mg %) due to
 - Inadequate dietary intake.
 - Hypoalbuminemia
 - Hypo parathyroidism
 - Renal disease/ failure
 - Vitamin D deficiency
- **Chronic deficiency** leads to loss of bone mass (bone resorption) and osteoporosis, bone fractures

PHOSPHORUS:

Important in Bones & teeth, Other
metabolisms rare Equal to calcium

SODIUM

- : Processed foods, salt Osmotic skeleton of extracellular fluid >10g intake/d increase B.P. 5g/d
- **Functions:**
- • Sodium maintains osmotic pressure of extra cellular fluid and ECF balance.
- • It has a role , along with others, in the neuro muscular excitability
- • Sodium is exchanged with Hydrogen in renal tubules to acidify urine.
- • Sodium pump keeps sodium in far higher concentration outside the cell .This results high polarization, create resting membrane potential.
- • Sodium and Potassium maintain the degree of hydration of plasma proteins, and there by viscosity of blood.
- • Potassium is critically important for the functioning of cardiac muscle

POTASSIUM

Occurs widely in foodstuffs, little likelihood of deficiency
Vasoactive, increases blood flow, sustains metabolic needs of tissue, lowers B.P., intracellular osmolality
Little likelihood of deficiency Hyperkalemia –ECG changes (cardiac arrest) 1:1 ratio of sodium: potassium (mmol)

Functions:

- • Constituent of bone and teeth
- • Needed for the synthesis of energy rich molecules like ATP and Creatin phosphate.
- • It forms Phosphate buffer in blood.
- • Constituent of phospholipids, biomolecules and coenzymes (TPP).

MAGNESIUM:

MAGNESIUM: Metabolism of calcium and potassium
Irritability, tetany, hyper-reflexia, hypo- reflexia
340mg/day

Functions:

- • Role in enzyme action. It is a cofactor for peptidases, ribonucleases, glycolytic enzymes
- • Its action is similar to that of calcium in neuromuscular irritability.
- • High levels depress nerve conduction, low levels may cause Tetany.
- • Major part is found in bones. In teeth, it is present as dentin and enamel.
- • Magnesium deficiency occurs rarely in man.